

UTH
May 26

①

$$\begin{aligned}\ddot{x} - z\dot{y} &= u_x \\ \ddot{y} + z\dot{x} &= u_y \\ \ddot{z} &= u_z\end{aligned}$$

$$\begin{aligned}u_x &= x - \frac{(1-\mu)(x+\mu)}{d_1^3} + \frac{\mu(1-\mu-x)}{d_2^3} \\ u_y &= y - \frac{(1-\mu)y}{d_1^3} - \frac{\mu y}{d_2^3} \\ u_z &= -\frac{(1-\mu)z}{d_1^3} - \frac{\mu z}{d_2^3}\end{aligned}$$

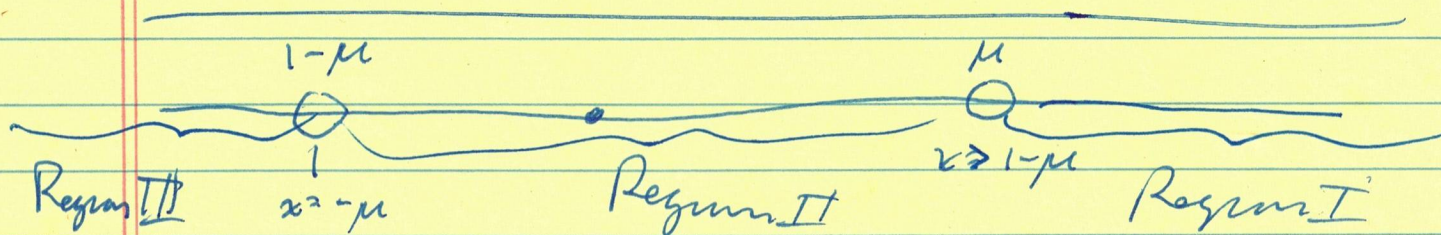
$$d_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}$$

$$d_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}$$

equilibrium points: $z = 0$

↳ colinear points: $y = 0$

$$x - \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} + \frac{\mu(1-\mu-x)}{|x-1+\mu|^3} = 0$$



Look @ Region II (note about Lagrange) where $x > 1-\mu$.

The others will look the same

$$|x+\mu| > 0 \Rightarrow (x+\mu) > 0$$

$$|x-1+\mu| > 0 \Rightarrow (x-1+\mu) > 0$$

Following ~~Szebehely's~~ ^{we need to define} Szebehely's convention define $\delta = x - 1 + \mu$

$$x+\mu = \delta+1 \quad x = \delta+1-\mu$$

$$\delta+1-\mu - \frac{(1-\mu)(\delta+1)}{(\delta+1)^3} - \frac{\mu(\delta)}{\delta^3} = 0$$

$$\cancel{(\delta+1-\mu)(\delta+1)^3} \\ \delta+1-\mu - \frac{(1-\mu)}{(\delta+1)^2} - \frac{\mu}{\delta^2} = 0 \quad \beta = 1-\mu$$

$$\begin{aligned} &(\delta+1-\mu)(\delta+1)^2\delta^2 - (1-\mu)\delta^2 - \mu(\delta+1)^2 = 0 \\ &(\delta+1-\mu)(\delta^2+2\delta+1)\delta^2 - (1-\mu)\delta^2 - \mu(\delta^2+2\delta+1) = 0 \\ &\cancel{(\delta^3+\delta^2-\mu\delta^2+2\delta^2+2\delta-2\mu\delta+\delta+1-\mu)\delta^2 - (1-\mu)} \end{aligned}$$

$$\delta + \beta - \frac{\beta}{(\delta+1)^2} - \frac{\mu}{\delta^2} = 0$$

$$\delta(\delta+1)^2 + \beta\delta^2 - \beta\delta^2 - \mu(\delta+1)^2 = 0$$

$$\cancel{\delta^3(\delta^2+2\delta+1) + \beta\delta^2(\delta^2+2\delta+1) - \beta\delta^2 - \mu}$$

$$\delta^3(\delta+1)^2 + \beta\delta^2(\delta+1)^2 - \beta\delta^2 - \mu(\delta+1)^2 = 0$$

$$\begin{aligned} &\delta^3(\delta+1)^2 + \beta\delta^2(\delta+1)^2 - (\beta\delta^2 + 2\beta\delta + \beta) + 2\beta\delta + \beta - \mu(\delta+1)^2 = 0 \\ &\cancel{(\delta+1)^2\delta^3 + \beta\delta^2(\delta+1)^2 - \beta(\delta+1)^2 + 2\beta\delta + \beta - \mu(\delta+1)^2 = 0} \end{aligned}$$

$$\cancel{\delta^3 + \beta(\delta^2 + 2\delta + 1) - \beta(\delta+1)^2 - \mu(\delta+1)^2 = 0}$$

$$\cancel{\delta^3 + \beta(\delta^2 + 2\delta + 1) - \beta(\delta+1)^2 - \mu(\delta+1)^2 = 0}$$

$$\cancel{\delta^3(\delta+1)^2 + \beta\delta^2(\delta+1)^2 - \beta(\delta+1)^2 + 2\beta\delta + \beta - \mu(\delta+1)^2 = 0}$$

$$\delta^3 \textcircled{1} (\delta+1)^2 + \beta\delta^2 \textcircled{2} (\delta+1)^2 - \beta(\delta+1)^2 \textcircled{3} + \beta(2\delta+1) - \mu(\delta+1)^2 \textcircled{4} = 0$$

$$(\delta^3 + \beta\delta^2 - \beta - \mu) \textcircled{1} \textcircled{2} \textcircled{3} \textcircled{4} (\delta+1)^2 + \beta(2\delta+1) = 0$$

$$(\delta^3 + \beta\delta^2 - \beta - \mu)(\delta^2 + 2\delta + 1) + \beta(2\delta + 1) = 0$$

$$\cancel{\delta^5 + \beta\delta^4 + 2\beta\delta^3 + \beta\delta^2 - \beta\delta^2 - 2\beta\delta - \beta - \mu\delta^2 - 2\mu\delta - \mu + 2\beta\delta + \beta = 0}$$

$$\textcircled{1} \delta^5 + \textcircled{2} \textcircled{3} \delta^4 + \textcircled{4} \textcircled{5} \delta^3 + \textcircled{6} \delta^2 - \textcircled{7} \delta^2 - \textcircled{8} \delta - \textcircled{9} \mu\delta^2 - \textcircled{10} 2\mu\delta - \textcircled{11} \mu + \textcircled{12} 2\beta\delta + \textcircled{13} \beta = 0$$

$$\delta^5 + (3+2\mu)\delta^4 + (2\mu+1)\delta^3 - \mu\delta^2 - 2\mu\delta - \mu = 0$$

$$\delta^5 + (3-\mu)\delta^4 + (3-2\mu)\delta^3 - \mu\delta^2 - 2\mu\delta - \mu = 0$$

check
against
Claude

quintic w/
no general
soln